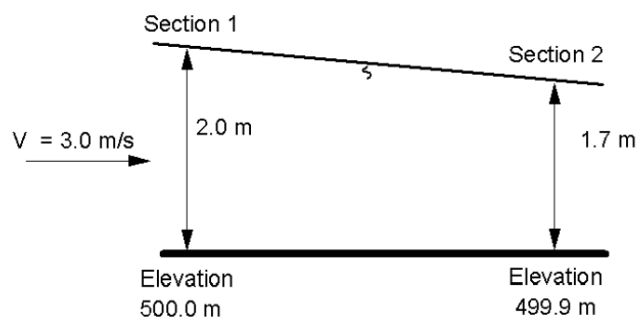


Topic	Surface Water Modelling	
Module	2.	Hydraulic Modelling (steady/unsteady)
Exercises		Solutions to Exercises

Exercise 1

The velocity of the upstream end of a rectangular channel 1 m wide is 3.0 m/s, and the flow depth is 2.0 m. The depth at the downstream end is 1.7 m. The elevation at section 1 is 500 m and at 2 is 499.90 m. Determine the head loss h_L due to friction. Assume the kinetic energy correction factor is 1.0.



Solution

Step 1: Use the continuity equation to find the discharge in the channel.

$$Q = V_1 A_1 = 3.0 (2.0) (1.0) = 6.0 \text{ m}^3/\text{s}$$

Step 2: Use the continuity equation to find the velocity at Section 2.

$$Q = V A$$

$$V_2 = Q / A_2 = 6.0 / [(1.7)(1.0)] = 3.53 \text{ m/s}$$

Step 3: Use the energy equation to find the headloss, h_L

$$\frac{V_1^2}{2g} + Y_1 + Z_1 = \frac{V_2^2}{2g} + Y_2 + Z_2 + h_L$$

$$\frac{3.0^2}{(2)(9.81)} + 2.0 + 500 = \frac{3.53^2}{(2)(9.81)} + 1.7 + 499.9 + h_L$$

$$h_L = (0.46 + 2 + 500) - (0.63 + 1.7 + 499.9) = 0.23 \text{ m}$$

Exercise 2

(a) A rectangular channel 5 m wide carries a discharge of 100 m³/s at normal depth. The channel is running on a slope of 0.005 m/m and has a Manning n value of 0.035 (metric). Determine the normal depth y_0 for this situation.

Solution

Employing

$$AR^{2/3} = (nQ)/S_0^{1/2}$$

With b = channel width and y_0 = normal depth

$$AR^{2/3} = \frac{nQ}{S_f^{1/2}}$$

Hydraulic radius $R=A/P$ (area divided by wetted perimeter)

Important: for uniform flow S_f (energy grade line slope) = S_0 (channel bottom slope)

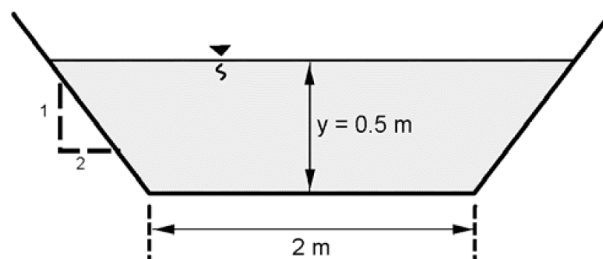
Also remember that Manning's equation was developed for uniform flow.

$$by_0 \left(\frac{by_0}{2y_0 + b} \right)^{2/3} = \frac{0.035 \times 100}{0.005^{1/2}} = 49.50$$

Use trial and error, spreadsheet software or write code to solve.

$$y_0 = 6.65 \text{ m}$$

(b) Determine the velocity (V) and discharge (Q) for the trapezoidal earth channel shown below, with $B = 2$ m, side slope 1V:2H, $S = 0.003$ m/m, normal (uniform) depth $y = 0.5$ m, and $n = 0.02$ (metric).



Solution

$$V = \frac{K_u}{n} R^{2/3} S^{1/2}$$

SI units; $K_u = 1$

$$R = \frac{By + Zy^2}{B + 2y\sqrt{1+Z^2}}$$

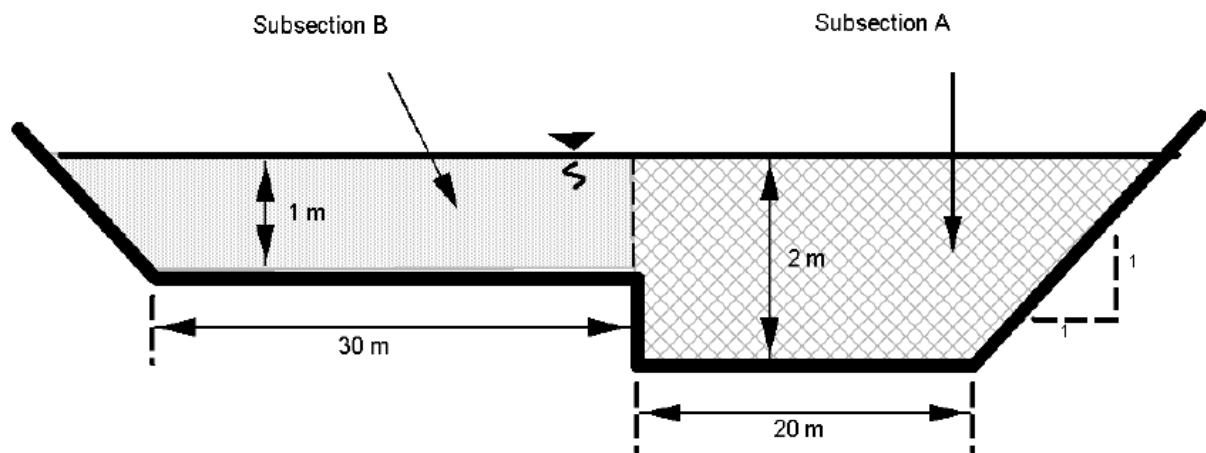
$$R = \frac{2(0.5) + 2(0.5)^2}{2 + 2(0.5)\sqrt{1+(2)^2}} = \frac{1.50}{4.24} = 0.35 \text{ m}$$

$$V = (1/0.02) (0.35^{2/3}) (0.003^{1/2}) = 1.36 \text{ m/s}$$

$$Q = V A = 1.36 (2(0.5) + 2(0.5)^2) = 2.0 \text{ m}^3/\text{s}$$

Exercise 3

A compound channel as illustrated, with an n value of 0.03, a longitudinal slope of 0.002 m/m and side slopes of 1V:1H.



Determine the discharge Q (i) with and (ii) without subdividing the channel and compare the results.

Calculate and plot the conveyance curve for this cross section and calculate the total discharge. Note: With a subdivided cross section, the conveyance at a given flow depth should be calculated for each subsection, and then added together to get the total conveyance at that depth.

Solution

(i)

Subsection A

$$\text{Area } A = (2)(20) + \frac{1}{2}(2)(2) = 42 \text{ m}^2$$

$$\text{Wetted perimeter } P = 20 + (2)(2)^{1/2} + 1 = 23.83 \text{ m}$$

$$\text{Hydraulic radius } R = 42 / 23.83 = 1.76 \text{ m}$$

$$\text{Flow velocity (using Manning's equation) } V = (1/0.03) (1.76)^{2/3} (0.002)^{1/2} = 2.17 \text{ m/s}$$

$$\text{Discharge } Q = 42(2.17) = 91.14 \text{ m}^3/\text{s}$$

Subsection B

$$A = 30(1) + (\frac{1}{2})(1)(1) = 30.5 \text{ m}^2$$

$$P = 30 + 1(2)^{1/2} = 31.41 \text{ m}$$

$$R = (30.5) / (31.41) = 0.97 \text{ m}$$

$$V = (1/0.03) (0.97)^{2/3} (0.002)^{1/2} = 1.46 \text{ m/s}$$

$$Q = 30.5 (1.46) = 44.53 \text{ m}^3/\text{s}$$

For the entire channel

$$A = 42 + 30.5 = 72.5 \text{ m}^2$$

$$Q = 91.14 + 44.53 = 135.67 \text{ m}^3/\text{s}$$

(ii)

If the channel had been considered as a whole without subdividing, the following results would have been obtained.

$$A = 42 + 30.5 = 72.5 \text{ m}^2$$

$$P = 23.82 + 31.41 = 55.23 \text{ m}$$

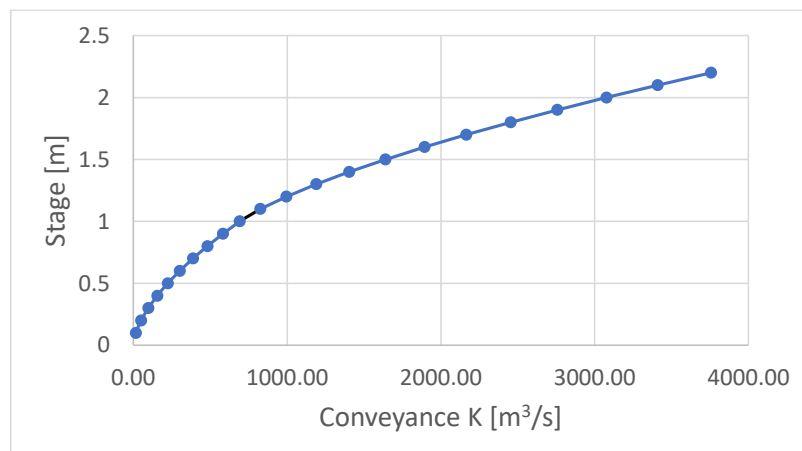
$$R = 72.5 / 55.23 = 1.31 \text{ m}$$

$$V = (1/0.03) (1.31)^{2/3} (0.002)^{1/2} = 1.78 \text{ m/s}$$

$$Q = 72.5 (1.78) = 129.05 \text{ m}^3/\text{s}$$

The resulting discharge using the whole channel is 4.9% less than the discharge obtained by subdividing the channel.

Conveyance calculation: The conveyance is plotted in the figure below.



At the main channel flow depth of 2 m, the total conveyance is 3037.1 m³/s. The total discharge is then

$$Q = K S^{1/2} = 3037.1 (0.002)^{1/2} = 135.8 \text{ m}^3/\text{s}$$

This result closely matches the discharge value for a 2 m flow depth for the case where the conveyance is calculated for the subdivided channel.

Exercise 4

Calculation of the water surface profile upstream of a dam

One purpose for which dams are designed and built is flood protection. Among the range of criteria that need to be considered, the design engineer must ascertain that the dam height is sufficient for the annual exceedance probability of the design flood, and the design engineer must also ascertain that the change in water level upstream of the dam does not result in negative consequences. You will investigate these two aspects in detail. For the channel under consideration with trapezoidal geometry (base width $b = 10$ m, side slope $1:m = 1:2$ and channel bed slope $S_0 = 0.00025$), compute the water surface profile for a design discharge $Q = 25$ m³/s and a Manning's roughness coefficient $n = 0.02857$ s/m^{1/3}. The flow is subcritical with a uniform flow depth $y_0 = 2.25$ m. The proposed dam blocks the water at $x = 0$ m up to an elevation $y_p = 382$ m above mean sea level with a water depth at the dam of $y = 3.10$ m.

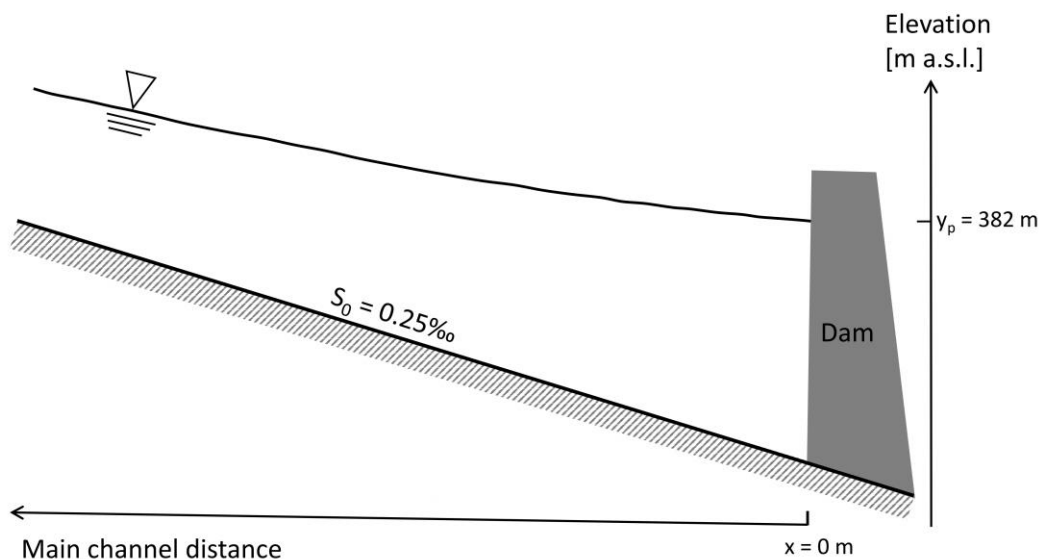


Figure 1a: Longitudinal-section view of the trapezoidal channel

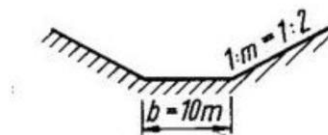


Figure 1b: Cross-section view of the trapezoidal channel

Determine the water surface profile using the standard step method.

Solution

See tutorial/computer lab for solution using HEC-RAS.